

02_METHOD

Method Justification Statement

1. Refined Research Question

Primary Research Question

How is digital transformation maturity associated with the quality and efficiency of higher education across organizational units at the Universidad Nacional Mayor de San Marcos, and what organizational and technological factors help explain these relationships?

Supporting Research Questions

1. Which dimensions of digital transformation maturity are associated with educational quality across organizational units at UNMSM?
2. How is digital transformation maturity associated with the efficiency of academic and administrative processes across organizational units at UNMSM?
3. Which organizational, technological, and human factors facilitate or hinder digital transformation at UNMSM?
4. What strategic actions can be proposed to strengthen digital transformation and support improvements in educational quality and institutional efficiency?

2. Candidate Research Methods

Method 1 – Quantitative Survey Research.

Method 2 – Qualitative Case Study.

Method 3 – Explanatory and Comparative Mixed-Methods Case Study.

3. Method-Fit Matrix

Criterion	Quant.	Qual.	Mixed	Score	Selected
Alignment with research question	4	4	5	5	✓
Ability to assess associations and differences	5	2	5	5	✓
Contextual understanding	2	5	5	5	✓
Practical applicability	4	4	5	5	✓
Methodological robustness	4	4	5	5	✓
Total	19	19	25	25	Mixed

4. Methodological Foundation

Creswell (2023): According to Creswell, mixed methods research is appropriate when complex problems require both quantitative measurement and qualitative explanation. Digital transformation involves technological, organizational, and educational dimensions simultaneously; therefore, integrating numerical evidence with contextual interpretation provides a more complete understanding.

Yin (2024): Yin argues that case study research is the preferred strategy for investigating contemporary phenomena within their real-world context. Since digital transformation is embedded in the institutional environment of UNMSM, the case study design allows an in-depth analysis of interactions among governance, technology, culture, and educational processes.

Saunders, Lewis & Thornhill (2023): The Research Onion framework proposes that methodological decisions should derive from the research objectives. The present study combines a pragmatic philosophy, an explanatory and comparative

case study strategy, and mixed methods to obtain rigorous and decision-oriented evidence.

Morgan (2024): Morgan identifies pragmatism as the philosophical foundation most compatible with mixed methods because it prioritizes the research question and the practical usefulness of knowledge. This perspective supports integrating multiple forms of evidence to solve institutional problems.

Tashakkori & Teddlie: These authors argue that triangulation and integration of quantitative and qualitative evidence increase credibility, validity, and explanatory power. Their framework supports combining surveys, interviews, institutional documents, and performance indicators in a coherent analytical design.

5. Operationalization of the Main Constructs

To ensure conceptual and analytical consistency, the three central constructs of the study - digital transformation maturity, educational quality, and institutional efficiency - will be operationalized through clearly defined dimensions, measurable indicators, survey items, institutional indicators, and corresponding analytical variables. This operationalization also links the research protocol to the reproducible data pipeline so that the constructs measured in the empirical study correspond to the features used in the analytical workflow.

Construct	Core dimensions	Indicators / measurement	Data source	Pipeline linkage
Digital Transformation Maturity	Digital infrastructure; digital governance; process automation; organizational digital capability	Scores derived from validated/selected survey items and institutional indicators for each dimension	Survey responses; institutional records and documents	digital_infrastructure_score; governance_score; automation_score; related DT features
Educational Quality	Teaching and learning quality; academic services; stakeholder experience	Quality indicators and survey-based measures defined in the final instrument	Student/faculty surveys; institutional quality indicators	Quality-related outcome variables/features defined in the data dictionary
Institutional Efficiency	Academic and administrative process efficiency; resource and service performance	Processing time, service performance, resource-use and survey-based efficiency indicators, as applicable	Administrative records; staff/user surveys; institutional indicators	Efficiency-related outcome variables/features defined in the data dictionary

Measurement traceability: Construct -> Dimension -> Indicator -> Survey/Institutional Measure -> Dataset Feature -> Analysis. The final protocol will identify the validated instrument or maturity model adopted for each construct and document the exact item-to-variable mapping in the data dictionary.

6. Unit of Analysis and Multilevel Structure

The study distinguishes between the unit of observation and the unit of analysis. Individual students, faculty members, administrative staff, and university managers constitute the units of observation because they provide survey and qualitative data. The principal comparative unit of analysis is the organizational unit within UNMSM, such as an academic or administrative unit, because the study compares units with different naturally occurring levels of digital transformation maturity.

Accordingly, the data have a nested or multilevel structure: individual respondents (Level 1) are nested within organizational units (Level 2), which are situated within the institutional context of UNMSM. Individual responses will not be treated as fully independent when organizational-level comparisons are made.

The quantitative analysis will account for this structure by examining within-unit and between-unit variation. Organizational-level aggregation will be used only when conceptually and statistically justified. If the number of organizational units and observations is sufficient, multilevel/hierarchical models will be considered; otherwise, clustered or unit-aware analytical procedures and sensitivity analyses will be used. The qualitative strand will complement these comparisons by examining contextual differences among organizational units.

7. Justification of the Selected Method

The Explanatory and Comparative Mixed-Methods Case Study provides the strongest alignment with the research objectives because it enables the assessment of associations between digital transformation maturity, educational quality, and institutional efficiency while using qualitative evidence to explain the organizational and technological factors underlying observed differences across organizational units. This design supports explanatory and comparative analysis without assuming a strong causal effect, and it ensures methodological rigor, contextual depth, and practical relevance for evidence-based decision-making at UNMSM. It also accommodates the multilevel structure of the study by distinguishing individual respondents as units of observation from organizational units as the principal comparative units of analysis.

8. Expected Methodological Contribution

The proposed methodology contributes a replicable framework for evaluating digital transformation in public universities. It integrates statistical evidence, organizational analysis, and stakeholder perspectives to generate recommendations that can strengthen educational quality and institutional efficiency in Peru and similar emerging contexts.